

Factoring Cubes and Higher Degrees

Factoring cubes and higher-degree expressions

Three patterns to recognise

What you see	Pattern	Factored form
Two terms, both cubes	$a^3 - b^3$	$(a - b)(a^2 + ab + b^2)$
Two terms, both cubes	$a^3 + b^3$	$(a + b)(a^2 - ab + b^2)$
Four terms	group in pairs	pull a common factor from each pair
x^4, x^2 , constant	quadratic in x^2	factor as if x^2 were a single letter

- Take out a common factor *first*, every time.
- *Factor completely*: keep going until no factor can be broken down further over the real numbers.
- The quadratic left over from a cube pattern ($a^2 \pm ab + b^2$) does not factor again — that is expected, not a dead end.

1. Factor completely: $x^3 - 27$

Answer: _____

2. Factor completely: $8x^3 + 125$

Answer: _____

3. Factor completely: $x^3 + 4x^2 - 9x - 36$

Answer: _____

4. Factor completely: $x^4 - 13x^2 + 36$

Answer: _____

5. Factor completely: $2x^3 - 16$

Look for a common factor before you reach for a pattern.

Answer: _____

6. Factor completely: $3x^3 - 2x^2 - 12x + 8$

Answer: _____

7. Consider $x^4 - 5x^2 + 4$.

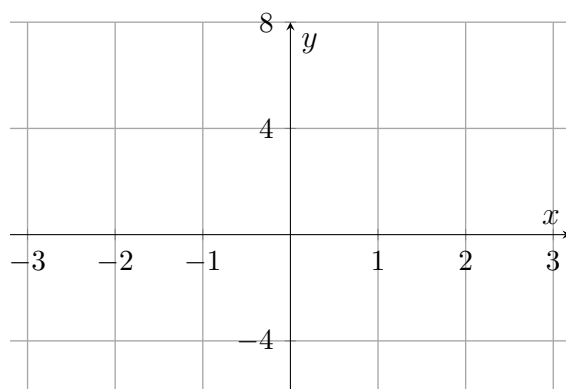
(a) Factor it completely.

Answer: _____

(b) Write the x -intercepts of the graph of $y = x^4 - 5x^2 + 4$.

Answer: _____

(c) Sketch the shape of the graph on the grid, marking those intercepts.



8. Factor completely: $64x^3 - 1$

Answer: _____

9. Factor completely: $x^6 - 64$

This is a difference of squares *and* a difference of cubes. Try the squares first and keep going.

Answer: _____

10. Consider $x^3 + 2x^2 - 4x - 8$.

(a) Factor it completely.

Answer: _____

(b) Solve $x^3 + 2x^2 - 4x - 8 = 0$ and list every solution.

Answer: _____

(c) One of those solutions is a zero the graph *touches* without crossing. Explain which one and why the factored form tells you so.

Answer: _____